Enrico Didoli

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Applied Mathematics, Statistics & Machine Learning

Profile

Applied Mathematician and Data Scientist with a background in economics, probability and statistical learning. Experienced in building reproducible machine and deep learning pipelines (Python, TensorFlow) for classification tasks on high-dimensional, imbalanced real-world data, with attention to calibration and uncertainty quantification. Interested in applying these methods to macroeconomic and policy analysis.

Education

2023-2026 **M.Sc. in Stochastics and Data Science | (Master's degree taught in English)**

University of Turin, Turin (Italy) | **Final Grade**: 105/110

- **Thesis**: "Detecting Measurable Residual Disease in B-cell Acute Lymphoblastic Leukaemia using Convolutional Neural Networks"
- **Supervisors**: Prof. Yvan Saeys, Prof. Francesca Romana Crucinio
- **Relevant coursework**: Econometrics, Simulation Models for Economics, Statistics for Stochastic Processes, Machine Learning
- **Links**: Thesis Repository | Thesis PDF

2024 **Erasmus+ Exchange Program | (Taught in English)**

Ghent University, Belgium (6 months study period)

- Brief introduction to PyTorch during a Machine Learning course taught by Prof. Saeys.

2020-2023 **B.Sc. in Economics and Data Science | (3rd Year taught in English)**

University of Turin, Turin (Italy) | **Final Grade**: 102/110

- **Thesis**: "Total variation between the law of the number of fixed points in a random permutation and its limit Poisson distribution"
- **Relevant coursework**: Microeconomics, Macroeconomics, Public Economics, Econometrics, Finance & Investments

Research Experience

2025–2026 **M.Sc. Thesis Researcher**

Department of Mathematics, University of Turin & Ghent University

Turin, Italy - Ghent, Belgium

- Extended and trained the CellCNN framework for Multiparameter Flow Cytometry sample classification (MRD detection) using TensorFlow.
- Designed patient-level data augmentation to prevent data leakage and improve generalization.
- Tuned classification thresholds and employed Bayesian optimisation to select internal CellCNN data-augmentation parameters on clinical datasets.
- **Key Results**: Recall = 0.95, F1 = 0.947 on held-out test set, with fully reproducible pipeline.
- **Ongoing extension**: CellCNN filter analysis for single-cell classification (clustering and filter selection, comparing logistic regression and mean-function thresholds). Repository

Projects

2026 **Independent project (in progress): Uncertainty-Aware RAG**

- Single-document RAG pipeline with a lightweight foundation model (Qwen-0.5B); retrieval and generation hyperparameters tuned with Bayesian optimisation (Optuna), maximising a lower confidence bound of semantic similarity.
- Calibration of a rejection threshold to limit out-of-context answers.
- Link: Project Repository

2024 **Calibration of ML Models Across Domains**, *Ghent University*

- Compared Temperature Scaling and Isotonic Regression for model calibration on CIFAR-10, part of coursework during my Erasmus+ semester.
- Found that calibration effectiveness is model- and domain-specific, giving me direct hands-on experience quantifying and validating model uncertainty.
- Link: Project Repository

2025 **Maze Generation Algorithms: Computational Complexity**, *University of Turin*

- Simulation study of five maze-generation algorithms (4500+ mazes each), analysing solve time, decision points and visited cells with an A* solver. Repository

Professional Background

2018-2021 Pastry Chef and Chocolate Artisan (Luxury Hospitality, Italy)

- Worked in high-pressure, precision-demanding environments before transitioning to academia to pursue applied mathematics. Cultivated adaptability, resilience, and strict attention to detail.

2017 Pastry Chef (Finland)

- Two-month internship around Finland. Cultivated adaptability, resilience, and teamwork skills.

Skills

Deep Learning	Computer Vision (CNNs, Transfer Learning), Foundation Models (Qwen), TensorFlow/Keras
Machine Learning & Uncertainty	Scikit-learn, Optuna (Bayesian Hyperparameter Optimisation), Imbalanced-learn, model calibration, conformal prediction
Data & Econometrics	Pandas, NumPy, Eurostat / ECB Data Portal (API), econometric modelling, simulation methods, flow cytometry analysis, handling clinical dataset variability
Visualisation & Office	Matplotlib, Seaborn, MS Excel, MS PowerPoint
Programming & Tools	Python, Git/GitHub, Jupyter, LaTeX, SQL (basic), Linux (WSL, basic)
Languages	English (B2.2; all M.Sc. and Erasmus coursework in English), Italian (native), French (A2)